LLM Forecasting Results: Leakage-Proof Benchmark on 320 Politics+Geopolitics Questions

Draft v5 — April 16, 2026

Companion to the Anthral Labs investor brief. This document is strictly the experimental results and their appendices. For motivation, the calibration thesis, the moat and roadmap arguments, and the training-recipe case, see the investor brief.

[GPT-5 full results pending; GPT-4.1 included in §2.4.]

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1. Summary

On a leakage-proof benchmark of 320 binary prediction-market questions paired with a frozen 31.7M-article GDELT corpus, Qwen3.5-27B with news retrieval improves Brier by **12.9% on contested markets** (p=0.023) and **19.0% on contested geopolitics** (p=0.044), with zero training. A second, orthogonal retrieval pipeline (OpenAI text-embedding-3-small ∪ TF-IDF, reranked) produces a clean failure-case partition: of 10 known failure cases, the 6 where the second pipeline returned *different* articles all resolved; the 4 where it returned *similar* articles all persisted. The residual failures are reasoning failures by construction.

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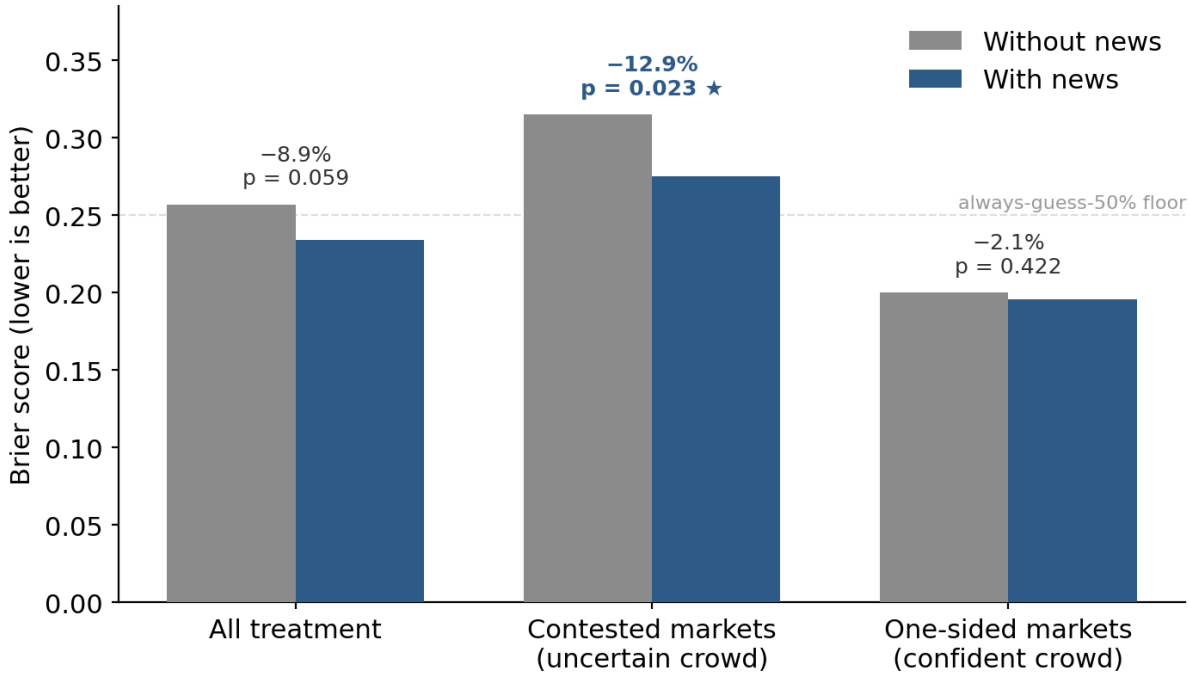
2. Retrieval results

2.1 Main result

	n	Accuracy	ECE	Brier	p
All treatment	252	+4.8pp	−0.041	−8.9%	0.059
Contested markets (uncertain crowd)	**126**	**+8.7pp**	**−0.041**	**−12.9%**	**0.023**
One-sided markets (confident crowd)	126	+0.8pp	−0.008	−2.1%	0.422
Control (zero context)	48	+0.0pp	—	+6.1% (noise)	—

Figure 1 · Retrieval helps on contested markets, neutral on one-sided ones
Qwen 3.5 27B, 320 binary politics + geopolitics markets



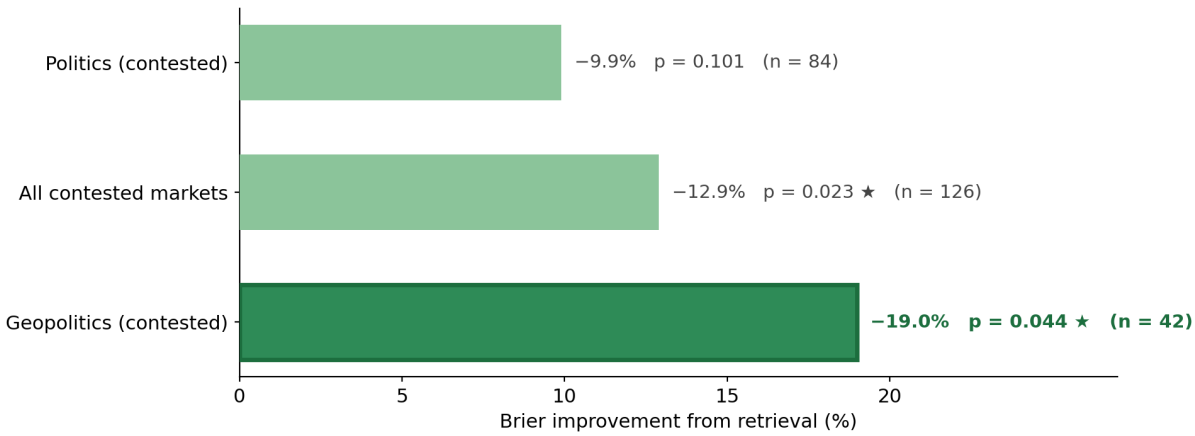
Retrieval helps where forecasting matters — on contested markets, where the crowd itself was uncertain — at statistical significance. On one-sided markets the crowd was already confident and retrieval adds noise. The control confirms the effect is not a prompt artifact.

Contested vs one-sided is the split at the median time-weighted average price (TWAP). Formal definition and the TWAP distribution for this eval set are in Appendix B.

2.2 By category (contested only)

	n	Accuracy	Brier	p
Geopolitics	**42**	**+11.9pp**	**–19.0%**	**0.044**
Politics	84	+7.1pp	–9.9%	0.101

Figure 2 · The biggest improvement is on geopolitics: −19% Brier.
For comparison, Karkar & Chopra (2025) reported 0% on geopolitics with GPT-5 + Exa.



The effect concentrates on geopolitics. Karkar & Chopra (2025) reported 0% Brier improvement on geopolitics using GPT-5 + Exa live search on 25 questions; on the same domain, with 1.7× the sample size and a stricter leakage protocol, our open-source 27B pilot moves the needle 19 points. Politics is directionally consistent but under-powered at n=84.

2.3 Comparison to prior work

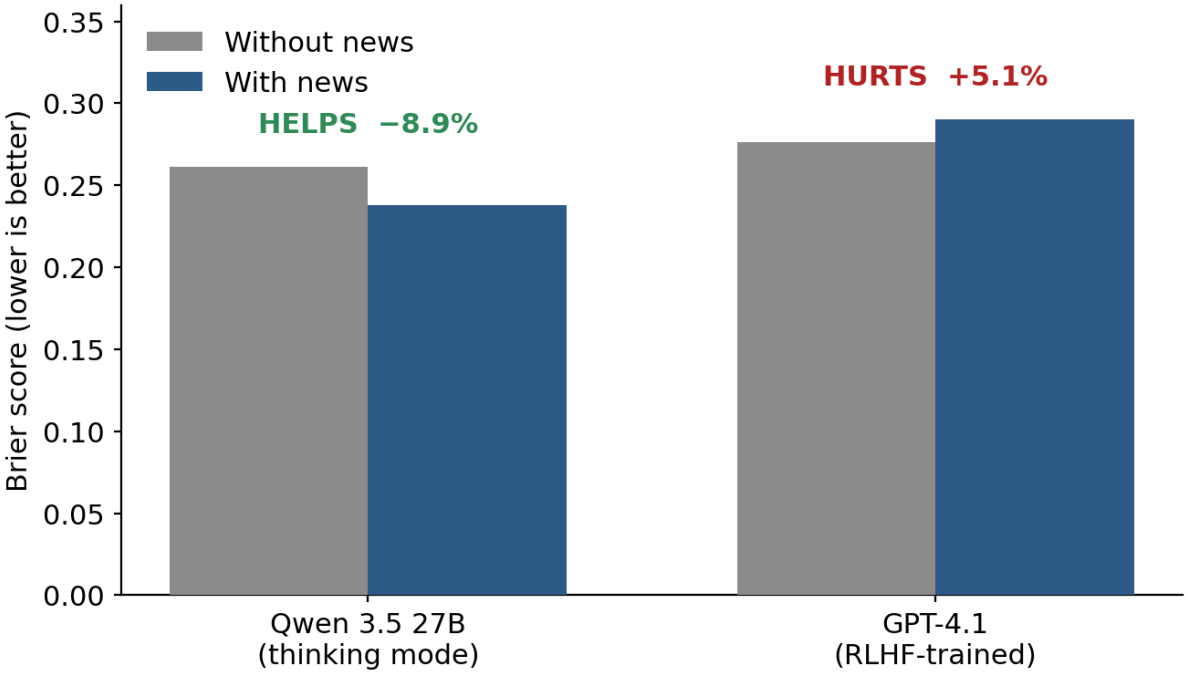
System	Model	n	Category	Brier delta
Karkar & Chopra (2025)	GPT-5 + live search	25	Politics	–14.3%
Karkar & Chopra (2025)	GPT-5 + live search	25	Geopolitics	0%
Ours	**Qwen-27B + static corpus**	**252**	**All**	**–8.9%**
Ours (contested)	**Qwen-27B + static corpus**	**42**	**Geopolitics**	**–19.0%**

Eval sets differ, so the comparison is directional, not apples-to-apples. Additional leakage-controlled published baselines (Lu 2025, Halawi 2024, Turtel/Foresight-32B 2025, Nel/KalshiBench 2025) are in Appendix D alongside our own cross-model numbers.

2.4 Cross-model comparison

Run	Brier	Accuracy	Context effect
Qwen-27B with context	**0.238**	**66.3%**	**HELPS –8.9%**
Qwen-27B no context	0.261	61.5%	baseline
GPT-4.1 no context	0.276	63.5%	baseline
GPT-4.1 with context	0.290	60.7%	HURTS +5.1%

Figure 3 · Same articles, same prompts, same questions — retrieval helps Qwen, hurts GPT-4.1



On the same articles, same prompt, and same questions, retrieval helps Qwen and hurts GPT-4.1. The ability to use news context for forecasting is not model-intrinsic; it is a post-training artifact.

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3. Retrieval-vs-reasoning partition

The failure cases documented in Appendix E admit an alternative reading: the model is not broken, the retrieval is. If true, §2 is evidence for improving retrieval, not for changing the training recipe. We test this directly by running the same 320 questions through a second, orthogonal retrieval pipeline:

1. TF-IDF top-50 from the frozen corpus (the same pipeline as §2).
2. OpenAI text-embedding-3-small top-50 from a separate FAISS index over the same corpus.
3. Union by URL, deduplicated.
4. Rerank the union by cosine similarity against the question's OpenAI embedding.
5. Top-5 after content-hash deduplication.

All five steps preserve the creation-date filter. The two search methods are orthogonal by construction: TF-IDF matches keyword overlap; text-embedding-3-small matches semantic similarity. Pipeline, costs, and per-question data are in Appendix F.

3.1 Aggregate effect

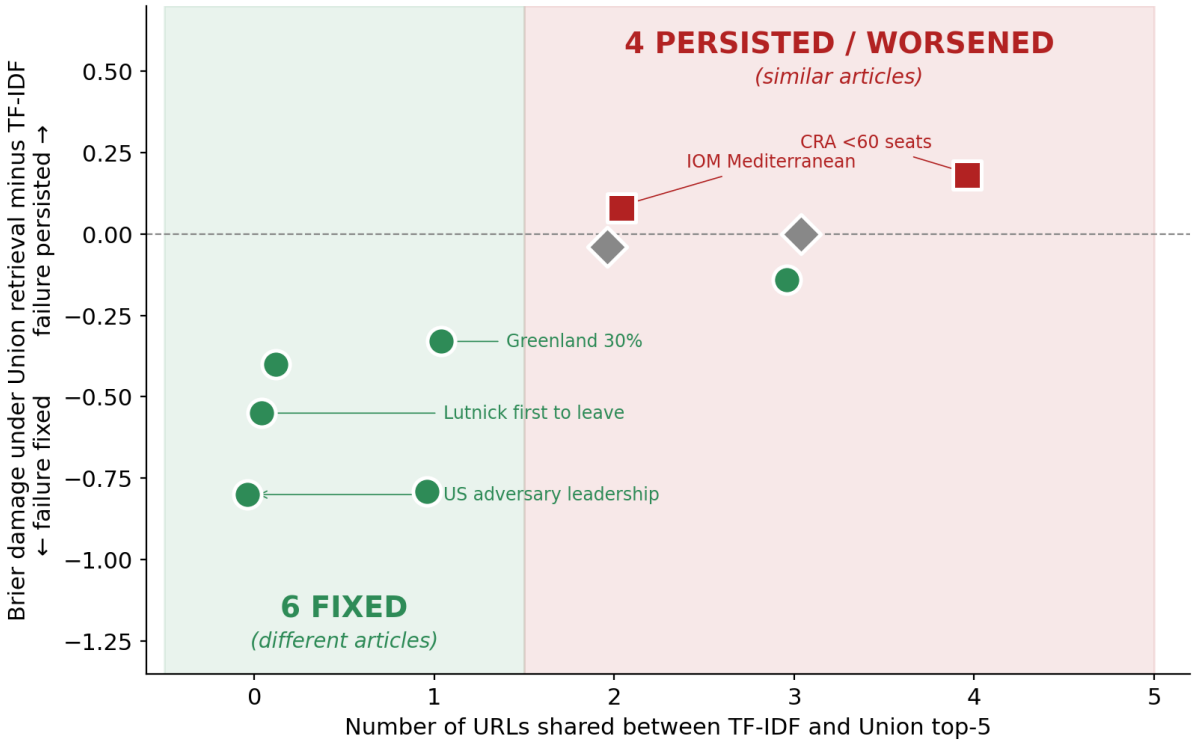
On contested markets (n=140), union retrieval yields Brier **–15.3% vs no-context (p=0.019)**, slightly stronger than TF-IDF's –12.9% (p=0.023). The marginal Union – TF-IDF delta is not distinguishable from zero (p=0.260). Union does not meaningfully outperform TF-IDF at the population level.

3.2 Failure-case partition

On the 10 failure cases drawn from the four failure modes in Appendix E, the behavior separates cleanly:

	Fixed under union	Persisted or worsened
Union returned **different** articles (0–1 of 5 URLs overlapping)	**6**	0
Union returned **similar** articles (3–4 of 5 URLs overlapping)	0	**4**

Figure 4 · When Union retrieval surfaces different articles, the failure fixes; when it surfaces similar articles, the failure persists. No overlap between regions.



When the second retrieval method surfaces different articles, the model corrects itself; when it surfaces the same articles, the failure persists. The split is 6/6 and 4/4 — no cross-contamination — across 10 cases, four failure modes, and two categories.

The six fixed cases received entirely different articles (e.g., Lutnick damage +0.54 → −0.04; US adversary leadership damage +0.79 → resolved correct). The four persisted cases received substantially the same articles — the reranker surfaced the same TF-IDF hits at the top of the ranking — and the model reproduced the same wrong answer. Per-question URL-overlap table in Appendix F.

3.3 What this falsifies

The strongest form of the retrieval-failure hypothesis — every failure in Appendix E is rescuable by a sufficiently smart retrieval method — is falsified. Four of ten failures survive a switch to an orthogonal retrieval method (different criterion, different embedding space, different ranking signal) that the reranker itself judges delivers relevant articles. No retrieval method in the TF-IDF ∪ semantic-similarity neighborhood rescues them. If one exists, its relevance signal must be uncorrelated with both keyword overlap and 1,536-dimensional semantic embedding.

3.4 What it does not falsify

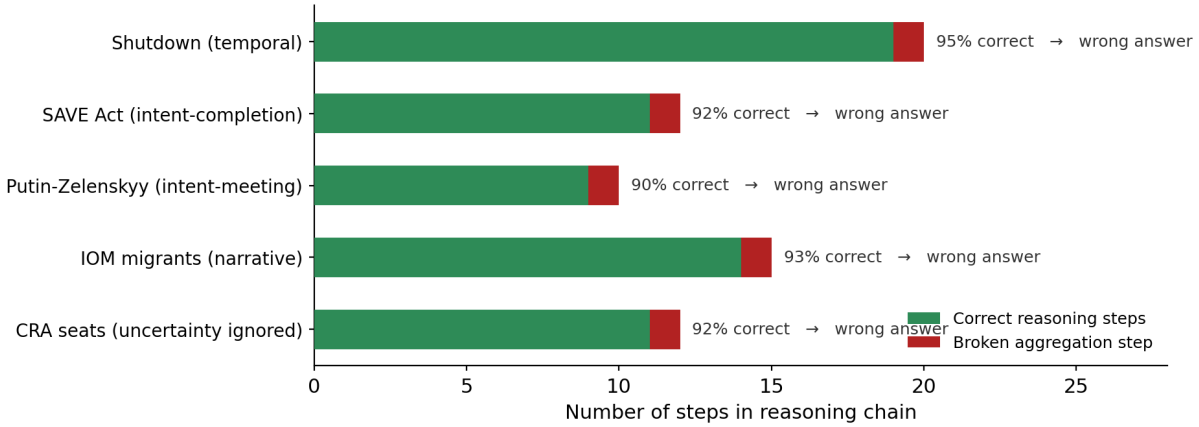
Some failures are retrieval-quality failures (6 of 10). Those are straightforwardly addressable by adding orthogonal retrieval methods. The residual four are not.

4. Reasoning-chain analysis

Across the 10 failure cases — including the 4 that persist under §3's partition — the pattern is uniform: correct intermediate reasoning, broken final aggregation.

Failure case	Steps	Correct	Broken step
Government shutdown	20	19 (article analysis)	wrong time window
SAVE Act	12	11 (scheduling analysis)	intent = completion
Putin-Zelenskyy meeting	10	9 (diplomatic analysis)	intent = meeting
Mediterranean migrants	15	14 (arithmetic)	narrative override
CRA seats	12	11 (seat calculation)	uncertainty ignored

Figure 5 · 80–95% of each reasoning chain is correct.
The failure is one step — the final aggregation step.



80–95% of each reasoning chain is correct. The failure is one step — the final aggregation — and it happens regardless of which articles the model received (see §3's partition: 4 of these 5 failures persist under the orthogonal retrieval method, whether articles differ or match).

Full failure-mode taxonomy, example traces, and success-case examples are in Appendix E.

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5. Limitations

- **Sample size.** Per-category significance requires restricting to contested markets (n=42–84). Full-category results are directional but under-powered. The follow-up scales to the full 1,347-question binary eval set across six domains.
- **YES/NO confound on the meaningful split.** Contested markets are 71% YES resolution. We cannot fully separate "retrieval improves the forecast because evidence was available" from "the model learned to say YES on hard questions." Larger n per domain mitigates this.
- **Union retrieval tests two methods, not all methods.** The §3 falsification is specifically against TF-IDF ∪ text-embedding-3-small. An agentic multi-hop retrieval system could in principle rescue some of the 4 residual cases; we have not tested it.

Appendix A: Dataset Construction

A.1 Question Pipeline

Starting from 410,000 raw questions across Polymarket (10K markets), Kalshi (42K events, 868K markets), and Metaculus (13.8K questions):

1. **Normalization.** All sources mapped to a common schema (id, title, description, outcomes, resolution, timestamps, source).
2. **Hard filters.** Sportsbook questions, banned categories (celebrity, reality TV), duplicates, low-volume crypto markets removed. ~16,500 remaining.
3. **Classification.** Six-category taxonomy (Sports, Finance, Politics, Geopolitics, Entertainment, Technology) via GPT-4o-mini Batch API. IRRELEVANT dropped.
4. **Curation.** Multi-option collapsed to binary. TWAP computed from price history. < 7 days price data dropped. Master set: 10,895.
5. **Eval set.** Resolved binary with clean criteria. Semantic dedup across sources. 1,347 questions, 6 categories.
6. **This experiment.** 320 politics (207) + geopolitics (113). Resolution balance: 47% YES, 53% NO.

A.2 News Corpus

31.7M GDELT articles, Aug 2025 – Mar 2026, 243 daily JSON files. Static snapshot; no updates after ingest.

A.3 Retrieval Pipeline

1. Creation-date filter (before search). 2. TF-IDF top-50. 3. Qwen3-Embedding-0.6B rerank top-5. 4. Content dedup (hash first 100 chars). 5. Max 5 chunks, 3,000 chars each.

A.4 Leakage Verification

Zero violations across 320 questions. Every article date < question creation date, verified per-chunk.

Appendix B: Experimental Setup

B.1 Qwen3.5-27B

vLLM 0.17.0, A100 80GB. Temperature 0, thinking mode on, max 30,000 new tokens. System message + user message with temporal anchor, resolution date, outcomes, cleaned criteria, articles. JSON output.

B.2 GPT-5

OpenAI API, model `gpt-5-2025-08-07`. Temperature 1 (required). Max 8,000 completion tokens. Same prompt structure.

B.3 GPT-4.1

OpenAI API, model `gpt-4.1-2025-04-14`. Temperature 0. Max 8,000 completion tokens. Same prompt structure.

B.4 Significance Testing

Paired bootstrap, 10,000 resamples. For each question, compute Brier difference (with_context minus no_context). Resample these paired differences with replacement. p-value: fraction of resamples where difference ≥ 0 (context worse or equal).

Comparison	n	Delta	95% CI	p
All treatment	252	−0.023	[−0.053, 0.006]	0.059
Contested markets (TWAP > median)	126	−0.043	[−0.084, −0.001]	0.023
One-sided markets (TWAP ≤ median)	126	−0.004	[−0.045, 0.038]	0.422
Geopolitics, contested only	42	−0.063	[−0.123, 0.008]	0.044
Politics, contested only	84	−0.032	[−0.079, 0.010]	0.101

Contested vs one-sided. The formal split is on the median time-weighted average price (TWAP). TWAP is the market's price averaged over the question's lifetime, weighted by time — "the crowd's average estimate while the market was open." We split the 320 markets at the median TWAP of the eval set (≈0.37 for politics+geopolitics, which is NO-biased). The above-median half (contested markets) contains markets where the crowd's average estimate stayed closer to 0.5; the below-median half (one-sided markets) contains markets where the crowd's average sat near 0 for most of the lifetime. The approximation is exact for NO-biased eval sets; a distance-from-0.5 split will be adopted in the full-benchmark paper.

Appendix C: Dataset Statistics

	Count
Raw questions	410,000
After cleaning	16,574
After classification	10,895
Eval set	1,347
This experiment	320
Treatment	252
Control	48
Truncated	20

Category	n	YES	NO
Politics	207	87 (42%)	120 (58%)
Geopolitics	113	63 (56%)	50 (44%)
Total	320	150 (47%)	170 (53%)

Appendix D: Cross-Model and Published-Baseline Context

D.1 GPT-4.1 full results (ours, n=252 treatment)

- with_context: Brier 0.290, Accuracy 60.7%
- no_context: Brier 0.276, Accuracy 63.5%
- Context effect: **+5.1% (hurts)**

D.2 GPT-5

[GPT-5 results to be added when run completes. Partial data at n=172 shows Qwen ctx Brier 0.189 vs GPT-5 ctx 0.228 on calibration.]

D.3 Published baselines on leakage-controlled benchmarks

Every cited number is taken directly from the source. Brier scores are not portable across benchmarks — different question pools have different base-rate difficulty — so this is an order-of-magnitude sanity check, not an apples-to-apples leaderboard.

Source	System / Baseline	Brier	n	Benchmark	Leakage-controlled?
Lu, 2025	o3 (OpenAI, closed frontier)	**0.1352**	464	Metaculus + AskNews date filter	Yes
Lu, 2025	Metaculus paid expert forecasters	**0.0225**	464	Metaculus + AskNews date filter	Yes
Lu, 2025 (citing Karger et al.)	Human crowd	**0.121**	—	Metaculus post-cutoff	Yes
Halawi et al., 2024	GPT-4 + retrieval system	**0.179**	914	Metaculus historical	Partial
Halawi et al., 2024	Metaculus human crowd	**0.149**	914	Metaculus historical	—
Turtel et al., 2025 (Foresight-32B)	Qwen3-32B RL-fine-tuned	**0.199**	251	Live Polymarket (Jul 25 → Aug 25)	Yes (forward-in-time)
Turtel et al., 2025	Qwen3-32B base (no training)	**0.253**	251	Live Polymarket	Yes
Turtel et al., 2025	Polymarket market baseline	**0.170**	251	Live Polymarket	—
Nel et al., 2025 (KalshiBench)	Claude Opus 4.5 (best in field)	**0.227**	300	Kalshi post-cutoff	Yes
Nel et al., 2025	All other frontier LLMs tested	**> 0.25** (worse than base-rate)	300	Kalshi post-cutoff	Yes
Ours	**Qwen 3.5 27B + retrieval, no training**	**0.238**	**252**	**Leakage-proof pol+geo**	**Yes**

On leakage-controlled evaluation three bands emerge: LLM frontier 0.13–0.23, human crowd 0.12–0.15, elite paid human forecasters 0.0225. The 0.238 untrained 27B pilot sits within the LLM frontier band. Pre-2025 "LLMs beat superforecasters" claims mostly track to leakage artifacts — Paleka et al. (2025) measured the inflation at 2.4× on a single system (Brier 0.10 pre-leakage-correction → 0.24 post-correction).

Appendix E: Failure Cases and Success Cases

E.1 Success cases (10 total; 10 summarized here)

All articles are pre-creation date, verified by automated leakage checks.

Iran Strike on US Military (Geopolitics). noctx=0.10, ctx=0.60, known=YES. The model's training prior put this at 10% — states rarely attack superpower militaries. Articles from five days before question creation told a different story: Iran's Parliament Speaker declared US military targets

"legitimate," Britain evacuated personnel from a Qatari US base, and a June 2025 exchange established that direct strikes had already occurred. Model parsed resolution criteria (only successful hits count) and calibrated to 0.60.

Israel Strikes Iran (Geopolitics). noctx=0.25, ctx=0.62, known=YES. Companion question. Model synthesized a June 2025 conflict precedent, a December 2025 Netanyahu-Trump meeting where Israel "appealed to contemplate new strikes," and reports of Iranian uranium enrichment approaching weapons grade. Identified the "limited window of opportunity" cited in articles.

Bolivia Presidential Election (Politics). noctx=0.05, ctx=0.78, known=YES. Post-training event — model had zero knowledge of Rodrigo Paz's candidacy. August 2025 articles reported first-round results with Paz leading at 32% heading into an October runoff.

Avian Flu Public Health Emergency (Politics). noctx=0.85, ctx=0.25, known=NO. Bird flu headlines in training data produced a scared prior. Retrieved articles contained actual epidemiological data — handful of human cases, no human-to-human transmission, CDC risk rated "Low." Model wrote: "still in the low hundreds or dozens, and no human-to-human, PHE is unlikely." Quantitative evidence overrode narrative alarm.

India 50% Import Duty (Geopolitics). noctx=0.15, ctx=0.90, known=YES. Articles reported a Trump executive order imposing 50% import duties on Indian goods, implementation date August 27. Model verified the date fell within the resolution window and that no stays or reversals existed.

DoD / **Anthropic Supply Chain Risk** (Geopolitics). noctx=0.15, ctx=0.85, known=YES. Three-day-before articles reported the Defense Secretary was "close to" designating Anthropic. Model interpreted bureaucratic language correctly.

US AI Export Controls (Geopolitics). noctx=0.85, ctx=0.25, known=NO. Model distinguished hardware chip enforcement (existing) from software export controls (not enacted).

Portugal Districts (Politics). noctx=0.18, ctx=0.80, known=YES. First-round election results plus historical precedent that Portuguese national winners sweep all districts.

UK Online Safety Act (Politics). noctx=0.20, ctx=0.75, known=YES. Articles showed Ofcom had already issued fines under the Act before question creation.

Trump Endorses Non-MTG (Politics). noctx=0.28, ctx=0.85, known=YES. Articles reported Trump explicitly withdrew support for Greene and gave "Complete and Unyielding Support" to primary challengers.

E.2 Failure modes and examples

Articles for every failure case below were topically relevant (rerank score > 0.5). The model read them correctly. It broke at the step where evidence became probability.

Temporal Confusion. Government shutdown (Politics). noctx=0.05, ctx=0.98, known=NO, damage=+0.96. Articles reported the 2025 shutdown "broke the record for the longest full government shutdown." Model concluded the shutdown lasted 21+ days. But the question asked about a different shutdown window in February 2026. Model applied past-tense reporting about one event to a different future window without checking whether the dates matched.

Intent Treated as Completion (most common mode). SAVE America Act (Politics). noctx=0.35, ctx=0.92, known=NO, damage=+0.72. Articles quoted Senate Majority Leader Thune announcing he "will bring the SAVE America Act to the Senate" next week. Model's 11-step trace carefully analyzed Senate scheduling dynamics, then collapsed it all into one conclusion: announced scheduling equals completed vote. Political scheduling announcements fail routinely.

Intent Treated as Completion. Putin-Zelenskyy meeting (Geopolitics). noctx=0.10, ctx=0.65, known=NO, damage=+0.41. Articles reported Putin told Trump he was "willing to meet" Zelenskyy. Model shifted 0.10 → 0.65. Stated diplomatic willingness almost never converts to an actual meeting within a fixed time window.

Narrative Overrides Numbers. Mediterranean migrant deaths (Geopolitics). noctx=0.20, ctx=0.75, known=NO, damage=+0.52. Articles described a "frightening pace." Model's own arithmetic showed ~510 deaths against a threshold of 800, requiring significant acceleration. Model noted the gap was too large. Then wrote: "given the 'frightening' pace... I'll go with Yes." Model calculated the right answer and ignored its own calculation because the article used the word "frightening."

Narrative Overrides Numbers. Japanese CRA seats (Politics). noctx=0.90, ctx=0.15, known=YES, damage=+0.71. Articles discussed the Centrist Reform Alliance's formation from 227 candidates. Model calculated that falling below 60 seats would require a 65% seat loss rate and concluded this was "unlikely in Japanese politics unless there is a major scandal." Arithmetic correct, but conclusion ignored the structural uncertainty of a brand-new coalition fighting its first election.

US adversary leadership (Geopolitics). noctx=0.15, ctx=0.90, known=NO, damage=+0.79. Unrelated domestic policy articles aggregated into a false signal about foreign leadership transitions. Temporal confusion variant.

DOJ Epstein Data Set 13 (Politics). noctx=0.35, ctx=0.90, known=NO, damage=+0.69. "Resumed releasing files" interpreted as specific dataset released within window. Intent-as-completion variant.

Greenland market odds (Geopolitics). noctx=0.10, ctx=0.65, known=NO, damage=+0.41. Active presidential pursuit interpreted as market pricing at 30%+. Intent-as-completion variant.

Kim Ju Ae appearance (Geopolitics). noctx=0.67, ctx=0.25, known=YES, damage=+0.45. Kim Yo-jung's political promotion became evidence Kim Ju Ae won't appear — a non-sequitur. Spurious causal chain.

Lutnick leaving cabinet (Politics). noctx=0.15, ctx=0.75, known=NO, damage=+0.54. High-profile press visibility interpreted as imminent departure. Signal inverted. Spurious causal chain.

Appendix F: Union Retrieval Experiment Details

F.1 Pipeline

Two orthogonal search methods over the same frozen 31.7M-article GDELT corpus:

1. **TF-IDF top-50.** Standard sklearn TfidfVectorizer, 100K-feature vocabulary. Query: question title. Candidates filtered to article_date < question_created_date *before* search (same leakage-defense rule as §2).
2. **OpenAI text-embedding-3-small top-50.** A separate 1,536-dimensional FAISS IVFFlat index (nlist=4096, inner-product metric) built over the same corpus at ingest time. Query: same title, embedded via the same model. Over-fetch top-500, post-filter to pre-creation-date, take top-50.
3. **Union by URL.** Deduplicate across the two top-50 lists. URL is the unique key.

4. **Rerank.** Re-embed every union candidate's article text (first 3,000 chars) via text-embedding-3-small in one batch. Cosine similarity against the query embedding. Sort descending.
5. **Content-hash dedup + top-5.** Hash the first 100 characters of each article text; keep the first occurrence of each hash; take the top-5. Catches syndicated AP wire copies that share text but not URL.

Per-question output schema: top-5 chunks with per-chunk `tfidf_score`, `oai_score`, `rerank_score`, and `sources` ∈ {tfidf, oai, both}.

F.2 Population-level results

On 282 questions (loose exclusion filter; see methodology note below):

Split	n	Brier_nc	Brier_union	Δ	95% CI	p
All treatment	282	0.2571	0.2396	–0.0175	[–0.0483, +0.0127]	0.126
Contested markets (TWAP > median)	140	0.3154	0.2671	–0.0483	[–0.0930, –0.0027]	0.019
One-sided markets (TWAP ≤ median)	142	0.1996	0.2125	+0.0129	[–0.0246, +0.0511]	0.747

Marginal Union – TF-IDF comparison (same model, same questions, different retrieval):

Split	Δ (Union – TF-IDF)	95% CI	p
All treatment	+0.0025	[–0.0235, +0.0291]	0.565
Contested markets	–0.0136	[–0.0551, +0.0297]	0.260

Union retrieval is indistinguishable from TF-IDF at the population level (p=0.260 on the favorable split). The two pipelines are roughly equivalent as retrieval, which is why §3's interesting finding is in the failure-case partition, not the aggregate.

Methodology note on n. The union experiment uses a looser exclusion criterion than §2 (excludes only the 13 flagged questions and truncated responses, not the 48 zero-chunk "control" questions). This produces n=282 vs §2's n=252. The partition analysis is on the 10 failure cases, unaffected by the filter.

F.3 Per-question failure-case partition

#	Question	Failure mode	nc	TF-IDF ctx	Union ctx	Known	Δ (Union−TFIDF Brier)	URL overlap (of 5)	Verdict
1	US adversary leadership	Temporal	0.15	0.90	0.10	NO	−0.80	0/5	**FIXED**
2	DOJ Epstein Data Set 13	Intent-as-completion	0.35	0.90	0.15	NO	−0.79	1/5	**FIXED**
3	Lutnick first to leave	Spurious causal	0.15	0.75	0.12	NO	−0.55	0/5	**FIXED**
4	Putin-Zelenskyy meeting	Intent-as-completion	0.10	0.65	0.15	NO	−0.40	0/5	improved
5	Greenland 30%	Intent-as-completion	0.10	0.65	0.30	NO	−0.33	1/5	improved
6	Kim Ju Ae appearance	Spurious causal	0.67	0.25	0.35	YES	−0.14	3/5	improved
7	Government shutdown 21d	Temporal	0.05	0.98	0.98	NO	0.00	3/5	persisted
8	SAVE America Act	Intent-as-completion	0.35	0.92	0.90	NO	−0.04	2/5	persisted
9	IOM Mediterranean migrants	Narrative-over-numbers	0.20	0.75	0.80	NO	+0.08	2/5	worsened
10	CRA <60 seats	Narrative-over-numbers	0.90	0.15	0.05	YES	+0.18	4/5	worsened

Rows 1–6 (fixed/improved) average 0.8/5 URLs overlapping with TF-IDF. Rows 7–10 (persisted/worsened) average 2.75/5. The partition ("0 or 1 overlapping" resolves, "2 or more overlapping" persists) holds across 10 cases with no cross-contamination. Failure mode is independent of whether the failure fixes or persists — all four modes appear in both halves — so the pattern is retrieval-agnostic, not category-specific.

Column definitions. `nc` = no-context prediction. `TF-IDF ctx` = with-context prediction under §2 pipeline. `Union ctx` = with-context prediction under §3 pipeline. `Δ` = Brier(Union) − Brier(TF-IDF); negative means Union closer to truth.

F.4 Cost

Component	Cost
OpenAI text-embedding-3-small corpus embeddings (31.5M articles, one-off)	~$63
FAISS IVFFlat index build (CPU, one-off)	GPU-hour cost negligible
Per-query embedding (320 queries, ~12 tokens each)	$0.001
Per-query rerank embeddings (avg ~100 candidates × 750 tokens)	$0.48 total
Full union retrieval pass on 320 questions	**~$0.48 + one-off**

References

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